Brass wiring

No insulated wire. Every connection is a bare 1 mm brass rod, soldered point to point like a circuit sculpture, and the device gets 40 % thinner because the header pins go. The electronics do not change: same pins, same resistors, same tests.

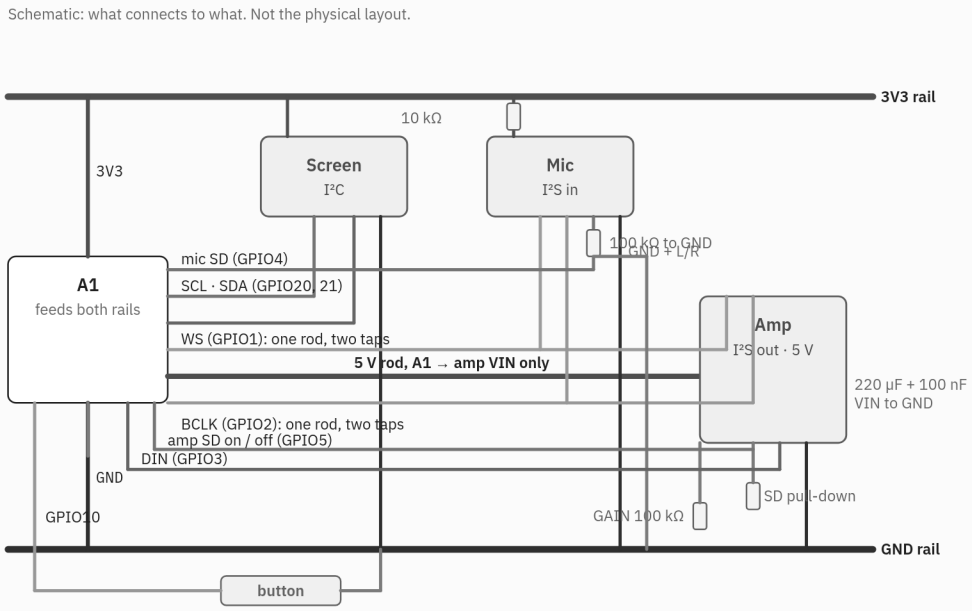
WHAT YOU NEED TO BUY

Nothing. The second pack of 1.0 mm rod you bought today (B005WPB7YG) covers all the wiring with rod to spare: about 300 mm of rod in total, one and a half sticks. Everything else is on your bench already (assessed below).

Where this fits

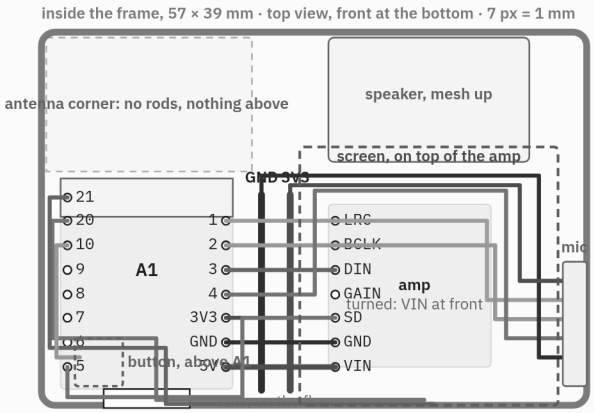
done	this week	Friday	after
Breadboard prototype	**Brass harness**	**Solder the frame**	**Into the frame**
the circuit works	rails and rods on the card, tested per module	posts now 18 mm, not 32	Kapton, place, glue, Wi-Fi check

The design: two rails, eleven rods



Two rails do most of the work. Nine of the thirteen connections touch GND or 3V3, so those become taps off two long rods. The rest are single point-to-point rods; the two clock rods each touch both the mic and the amp. Colours only mark the signal; every rod is bare brass.

Where every rod goes



Why it is this tidy

Turn the amp so VIN faces the front and its row lines up with A1's row: 5V–VIN, GND–GND, GPIO3–DIN, GPIO2–BCLK, GPIO1–LRC are five straight rods.

The rails lie on the floor; the straight rods pass 3 mm above them at stub height, so they never touch.

Rods from A1's left row (GPIO5, 10, 20, 21) run along A1's left edge and over its front half, never the back.

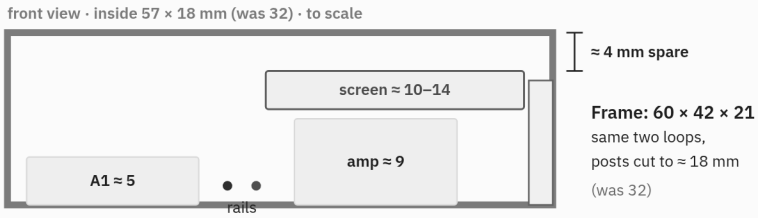
The two clock rods continue past the amp to the mic; the mic's taps come off the back ends of the rails.

Proposed placement, to scale. Rod colours only name the signal; every rod is bare brass. Lines are drawn flat; in the build the rods step up and down between the floor, the rails and the stub tops. Confirm the positions on the 57 × 39 card before bending a single rod.

ROD	COUNT	LENGTH	ROUTE
GND rail	1	≈ 22 mm	front to back in the channel, 2 mm above the floor
3V3 rail	1	≈ 22 mm	beside the GND rail, 3 mm apart
Rail taps	8	4–8 each mm	A1 GND, A1 3V3, amp GND, mic GND, mic VDD, screen GND, screen VCC, button GND
5 V rod	1	≈ 14 mm	A1 5V stub to amp VIN stub, straight
WS, BCLK	2	≈ 40 each mm	A1 → mic, then on to the amp: one rod, two taps each
mic SD, amp DIN, amp SD	3	12–38 mm	point to point
Screen SCL, SDA	2	≈ 30 mm	from A1's left row, up and over to the screen
Button rod	1	≈ 12 mm	GPIO10 stub up to the button

About twenty pieces of rod, 300 mm in all. Cut each one 10 mm long and trim after bending. The small parts stay the same as on the breadboard: the mic's 10 kΩ (SCK to 3V3 rail) and 100 kΩ (SD to GND rail), the amp's SD pull-down and 100 kΩ GAIN resistor to the GND rail, and the 220 µF + 100 nF across VIN and GND at the amp. Their own leads are the "rods" for those.

How compact it gets

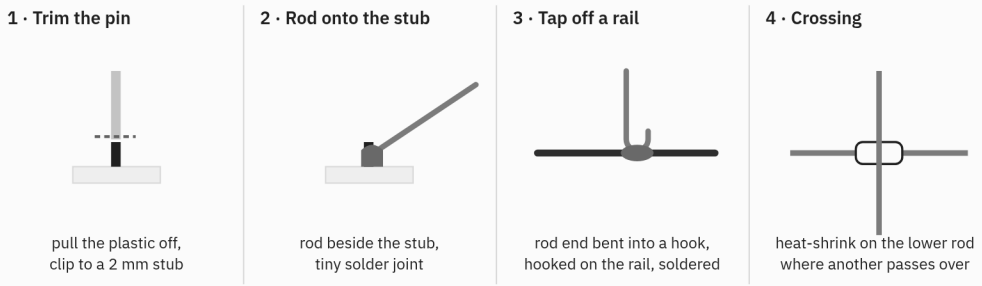


40 % thinner. Trimming the header pins to 2 mm stubs removes 8–10 mm from every module. Heights are estimates from your measurements; stack the real parts on the card and measure before cutting new posts.

> CHANGE TO FRIDAY'S FRAME
>
> Keep the two loops you bent: the footprint stays 57 × 39 mm. Cut **four new posts at about 18 mm** instead of 32 (measure your real stack first). The frame becomes **60 × 42 × 21 mm**. The four 33 mm posts you already cut become spare rod.

The four joints

Use: ROD 1.0 mm · K&S #9861 · B005WPB7YG · electronics tip · Chip Quik flux pen · 63/37 solder · USB unplugged



Four joints cover the whole build. Sand every rod end shiny, flux it with the Chip Quik pen, and use the electronics tip. Each joint takes 1–2 seconds; the modules must not be heated longer.

1. **Trim the headers on every module.** Grip the black plastic strip with the chain-nose pliers and pull it off the pins, then clip each pin to a 2 mm stub with the flush cutters. Do not desolder anything.

2. **Make the two rails.** Cut two 24 mm rods, sand them shiny along their whole length, and push them through a strip of cardstock 3 mm apart: the card is the jig that holds them straight and 2 mm off the floor.

3 **Place the modules on the 57 × 39 card** in the drawn positions, stubs facing the channel. Tape them down with Kapton.

4 **Rod by rod, in the build order on the drawing.** Bend first, hold with the helping hands, flux, one small joint each end, 1–2 seconds of iron per joint.

5 **After every rod, meter check:** the new rod beeps only to its own two ends; the GND rail never beeps to the 3V3 rail or to the 5 V rod.

6 **Sleeve every crossing** as you go, on the lower rod, before the upper rod is soldered.

7 **Resistors and capacitors: trim their leads.** They come 25–30 mm long; each lead is just the rod that reaches its two points. Measure the resistor with the meter first, hold the lead with the chain-nose pliers next to the body and bend beyond the pliers (never at the body), shape each lead to its stub or rail with a small hook, cut 3 mm past the joint with the flush cutters, solder, and sleeve any lead within 2 mm of another rod. The 6 mm body cannot shrink, so a resistor needs about 10 mm between its two points.

8 **Test per module**, in the same order and with the same tests as the wiring plan: button, screen, mic, amp, speaker, then two cold starts.

- [ ] **WIRE-01 · THE BRASS HARNESS WORKS OFF THE FRAME**
  All five modules pass on the rods, two cold starts are clean, every crossing is sleeved, and no rail beeps to the other rail. Tell Claude, with a photo, and send the measured stack height for the new posts.

Rules that do not change

- **The frame carries nothing.** The video solders its grounds to the cage; this build does not. Rails and rods stay at least 2 mm from the frame, with Kapton on the frame wherever they come close. A rail touching the frame makes every accidental contact a short, and a grounded frame sits as a ground plane beside A1's already-weak antenna.
- **Neither speaker output touches GND**, the GND rail or the frame.
- **5 V goes only to the amp.** The screen and mic get 3.3 V from the 3V3 rail.
- **Mic port:** no flux, heat gun or sanding dust near it.
- **USB unplugged** for every joint; plug in only to test.

Soldering kit assessed

PRODUCT YOU BOUGHT	VERDICT	NOTE	LINK
X-Tronic 3020-XTS station	✔ **right tool**	use a fine conical or small chisel tip for the stubs; keep the brass-frame tip separate	owned · no link recorded
MAIYUM 63/37 solder, 0.8 mm	✔ **fine**	thinner would be nicer on the stubs but is not needed	B076QF1Y85
Chip Quik CQ4LF no-clean flux pen	✔ **works on brass**	only if the rod end is sanded shiny first; never the Harris flux near a module	owned (Adafruit order)
Harris SCLF4 acid flux	✘ **frame only**	corrodes the modules; stays in the brass zone	B0015DWPV8
JoTownCand solder wick + X-Tronic solder sucker	✔ **needed**	clearing bridges between 2.54 mm stubs	B0DRN688Q5
WORKPRO chain-nose and round-nose pliers	✔ **needed**	bending hooks and corners in the rod	B0B8QBVXXR
BOENFU flush cutters	✔ **needed**	trimming header pins and rod ends (fine for 1 mm rod; never the tube)	owned · no link recorded
SE jeweler's saw	**optional**	only if a cutter leaves a crushed end	B06XPSLS6N
BAISDY sandpaper, 400 grit	✔ **needed**	every rod end shiny before flux	B07D2YYC11
Pointool heat-shrink, 1.5–2 mm sizes	✔ **needed**	sleeves at every crossing and near the frame	B08N4W4K9X
QWORK heat gun	✔ **needed**	shrinking sleeves; keep it off the mic port	owned · no link recorded
ELEGOO Kapton tape	✔ **needed**	lining the frame where rails and parts come near it	B072Z92QZ2
Helping hands, tweezers, silicone mat	✔ **needed**	holding rods at the right angle while the joint sets	owned
KAIWEETS multimeter	✔ **needed**	continuity after every rod: rail to frame must never beep	B08BL288LW
Amazon Basics cardstock	✔ **needed**	a jig: push rods through a card to hold them while soldering	B0DSH5V1TT

Trade-offs, honestly

BRASS RODS	INSULATED WIRE
thinner device, nothing to hide, looks like the reference build	bulkier, wires visible
every rod is bare: a bump can short two rods; sleeves and spacing matter	shorts almost impossible
stiff: the harness is its own structure, modules are held by their rods	floppy: needs hot glue to hold modules
rework means unsoldering a rod	rework is easy
about 2 sessions longer to build	faster

Sources. Connections, resistors and capacitors from `plan/FAST_TRACK.md` Steps 6–11 and the `CLAUDE.md` hardware contract; A1's pin order from the SuperMini pinout used on the Pin Switchboard page; part sizes from your caliper measurements of 28 September; the second rod pack from `docs/INVENTORY.md` (Amazon #16, 8 October). Product links are the listings on your order sheets.

What is and is not verified. This is a redesign on paper. Rod lengths, the placement and the 18 mm height are estimates to confirm on the card with the real parts before anything is cut or soldered. The frame-as-ground rule comes from the project's safety invariants. The checkbox is remembered only in this browser.